Error Analysis

Hard to show convergence → need FDE's to model physical properties/parameters

Conservation principle → In - out + gen = Acc. for a CV
"conservative property"

For fluid flow, a disturbance at any location in the flows field can spread to other locations by three transport mechanisms

- Convection (due to fluid motion)
- Diffusion
- Pressure waves

} FDE's have transportive property

spreads in every direction by random molecular motion → but disturbance is smoothed out by process called dissipation

If $\frac{\partial}{\partial t}$ (1st order in time), then even-order spatial derivatives are dissipation, i.e. $\mu \nabla^2 u$, $\mu \nabla^2 v$

truncation error w/ leading even order space derivatives are termed as dissipative terms

even likes to "diss"

Given a PDE, we can define multiple FDE's
↳ Now of the FDE's we can write modified equations (Taylor expansion)
↳ The order of dissipation of an FDE is the lowest even-order spatial derivative of the modified equation
↳ For ex: Pg 65, book
↳ A 4th order dissipation means we will be less dissipation error than that of a first order

Dispersion → solutions (= waves w/ Fourier Transform) that have different wavelengths travel at different speeds } speed ∝ f(wavelength)
↓
error if waves not travelling at same speed.
All fluids are modelled as non-dispersive

↳ presence of odd-order space-derivative in the TE is responsible for dispersion

When there is $\frac{\partial}{\partial t}$ term (first order) → even derivatives = diffusion error (dissipation)
odd derivatives = dispersion error

↓
∴ coefficients of FDE should be same as PDE for even
to model property
↓
If not then dissipative error